SONICCloud – Project Management Case Study

This document describes a comprehensive project management case study to be used by students during the course. The case study represents a realistic software project in an early planning phase, before detailed requirements elicitation.

1. Case Study Overview

Project name: SONICCloud
Domain: Cloud-based music streaming and sharing platform
Inspiration: SoundCloud (academic, non-commercial clone)

Project goal: The goal of the SONICCloud project is to design and implement a scalable, cloud-native music platform that enables independent artists to upload, manage, and share their audio content, while allowing listeners to stream music, follow artists, interact through likes and comments, and discover new tracks. The platform must support a large number of concurrent users, guarantee acceptable performance for audio streaming, and provide a modern and intuitive user experience across web and mobile devices.

2. Context and Assumptions

SONICCloud is commissioned by a start-up company aiming to enter the online music streaming market, focusing on independent musicians rather than major record labels. The system will be developed from scratch by a small-to-medium software team. At this stage of the project, only high-level assumptions are available; detailed functional and non-functional requirements will be defined in a later phase.

Key assumptions:

- The platform will be accessible via a web application and native mobile applications
- A cloud infrastructure will be used for storage, computation, and content delivery
- The system must scale to at least 100,000 registered users
- Audio content will be uploaded by users and streamed on demand
- The planned development duration is limited to 9 calendar months

3. Stakeholders

- Client organisation (SONICCloud Ltd.)
- End users: independent artists and listeners
- Development team (project manager, developers, designers, testers)
- External service providers (cloud and third-party services)

4. High-Level Product Description

The SONICCloud system consists of several interconnected components. Users interact with the system through web and mobile applications. Backend services manage user accounts, audio metadata, streaming requests, and social interactions. Audio files are stored and delivered using cloud-based storage and content delivery mechanisms. Administrative tools are provided to support content moderation and platform management.

5. Project Management Exercises

Exercise 1 – Product Breakdown Structure (PBS)

Define a Product Breakdown Structure for the SONICCloud system. Identify all major deliverables and decompose them into sub-products. Clearly distinguish between software components, infrastructure elements, and documentation artefacts.

Exercise 2 – Work Breakdown Structure (WBS)

Based on the PBS, define a Work Breakdown Structure for the project. Identify the main activities required to deliver each product component. Keep the WBS at a high level and do not assume detailed functional requirements.

Exercise 3 – Gantt Chart and Scheduling

Create a high-level Gantt chart for the SONICCloud project. Assign estimated durations to each major activity and identify dependencies between them. Determine the critical path of the project.

Exercise 4 – Risk Identification and Analysis

Identify at least five relevant risks related to the SONICCloud project. For each risk, specify its category, probability, impact, and overall risk exposure. Propose an appropriate mitigation or management strategy.

Exercise 5 – Preparation for Effort Estimation

Explain why a quantitative effort estimation using COCOMO II cannot be performed at this stage of the project. Identify the information that will be required after requirements elicitation to enable a reliable effort and schedule estimation.